An implicit rank-adaptive method for solving high-dimensional time-dependent problems using the hierarchical Tucker (HT) decomposition

Paolo Bosques-Paulet (Swarthmore College)

Joseph Nakao (Swarthmore College)
Gianluca Ceruti (formerly University of Innsbruck)

2026 SIAM NNP Section Conference

Outline

1. Motivation and hierarchical Tucker representation
2. The first-order HT-RAIL algorithm
3. High-order time integration
4. Numerical experiments
5. Closing remarks

Outline

1. Motivation and hierarchical Tucker representation
2. The first-order HT-RAIL algorithm
3. High-order time integration
4. Numerical experiments
5. Closing remarks

Solving high-dimensional diffusion equations

Consider the anisotropic diagonal diffusion equation of dimension $d \geq 4$,

$$u_t = \sum_{i=1}^{d} d_i(t) u_{x_i x_i}, \qquad \mathbf{x} \in \Omega \subset \mathbb{R}^d. \tag{1}$$

- Applications include stochastic processes, uncertainty quantification, PDE-constrained optimization, and kinetic theory.
- Rapid changes in short-time diffusive dynamics create stiffness
- With N grid points per dimension, storing the full solution requires $\mathcal{O}(N^d)$ entries.

Goal: Handle stiffness and reduce computational costs

Solving high-dimensional diffusion equations

Consider the anisotropic diagonal diffusion equation of dimension $d \geq 4$,

$$u_t = \sum_{i=1}^{d} d_i(t) u_{x_i x_i}, \qquad \mathbf{x} \in \Omega \subset \mathbb{R}^d. \tag{1}$$

- Applications include stochastic processes, uncertainty quantification, PDE-constrained optimization, and kinetic theory.
- Rapid changes in short-time diffusive dynamics create stiffness
- With N grid points per dimension, storing the full solution requires $\mathcal{O}(N^d)$ entries.

Goal: Handle stiffness and reduce computational costs

- **Implicit integration:** address stiffness from diffusion.
- **Low-rank storage:** exploit compressible structure in the solution.
- **Rank adaptivity:** adjust the representation as the solution evolves.

Reduced Augmentation Implicit Low-rank (RAIL)
in the hierarchical Tucker format.

The balanced hierarchical Tucker representation ($d = 4$)

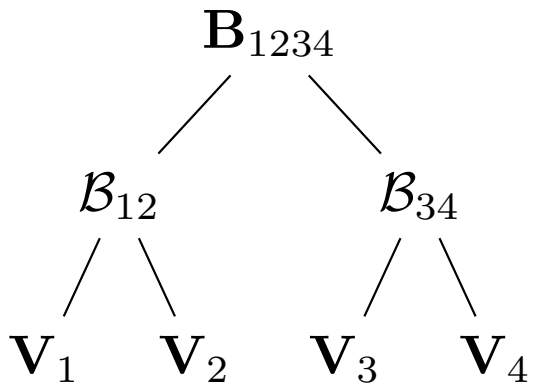
Lemma

Let $\mathcal{U} \in \mathbb{R}^{I_1 \times I_2 \times \ldots \times I_d}$ *and* $\alpha = \alpha_l \dot{\cup} \alpha_r$, *where* $\alpha = \{l, ..., r\} \subset \{1, ..., d\}$ *is the parent node and the children nodes are* $\alpha_l = \{l, ..., m\}$ *and* $\alpha_r = \{m+1, ..., r\}$. *Then,*

$$\operatorname{span}(\operatorname{mat}_\alpha(\mathcal{U})) \subset \operatorname{span}(\operatorname{mat}_{\alpha_r}(\mathcal{U}) \otimes \operatorname{mat}_{\alpha_l}(\mathcal{U})). \tag{2}$$

Implication: For some core tensor $\mathbf{B}_\alpha \in \mathbb{R}^{r_{\alpha_l} r_{\alpha_r} \times r_\alpha}$,

$$\operatorname{mat}_\alpha(\mathcal{U}) = (\operatorname{mat}_{\alpha_r}(\mathcal{U}) \otimes \operatorname{mat}_{\alpha_l}(\mathcal{U}))\mathbf{B}_\alpha$$



$$\mathbf{V}_i \in \mathbb{R}^{N_i \times r_i},$$
$$\mathcal{B}_{12} \in \mathbb{R}^{r_1 \times r_2 \times r_{12}},$$
$$\mathcal{B}_{34} \in \mathbb{R}^{r_3 \times r_4 \times r_{34}},$$
$$\mathbf{B}_{1234} \in \mathbb{R}^{r_{12} \times r_{34}}.$$

The balanced hierarchical Tucker representation ($d = 4$)

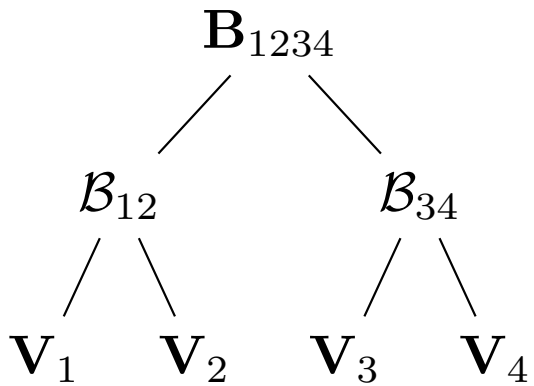
Lemma

Let $\mathcal{U} \in \mathbb{R}^{I_1 \times I_2 \times \ldots \times I_d}$ *and* $\alpha = \alpha_l \dot{\cup} \alpha_r$, *where* $\alpha = \{l, \ldots, r\} \subset \{1, \ldots, d\}$ *is the parent node and the children nodes are* $\alpha_l = \{l, \ldots, m\}$ *and* $\alpha_r = \{m+1, \ldots, r\}$. *Then,*

$$\operatorname{span}(\operatorname{mat}_\alpha(\mathcal{U})) \subset \operatorname{span}(\operatorname{mat}_{\alpha_r}(\mathcal{U}) \otimes \operatorname{mat}_{\alpha_l}(\mathcal{U})). \tag{2}$$

Implication: For some core tensor $\mathbf{B}_\alpha \in \mathbb{R}^{r_{\alpha_l} r_{\alpha_r} \times r_\alpha}$,

$$\operatorname{mat}_\alpha(\mathcal{U}) = (\operatorname{mat}_{\alpha_r}(\mathcal{U}) \otimes \operatorname{mat}_{\alpha_l}(\mathcal{U}))\mathbf{B}_\alpha$$



$$\mathbf{V}_i \in \mathbb{R}^{N_i \times r_i},$$
$$\mathcal{B}_{12} \in \mathbb{R}^{r_1 \times r_2 \times r_{12}},$$
$$\mathcal{B}_{34} \in \mathbb{R}^{r_3 \times r_4 \times r_{34}},$$
$$\mathbf{B}_{1234} \in \mathbb{R}^{r_{12} \times r_{34}}.$$

Representation	Storage
Full tensor	$\mathcal{O}(N^d)$
Tucker	$\mathcal{O}(dNr + r^d)$
Hierarchical Tucker	$\mathcal{O}(dNr + (d-2)r^3)$

Refolding the solution locally

Based on equation (2),

$$\mathrm{vec}(\mathcal{U}) = (\mathbf{V}_4 \otimes \mathbf{V}_3 \otimes \mathbf{V}_2 \otimes \mathbf{V}_1)(\mathbf{B}_{34} \otimes \mathbf{B}_{12})\,\mathrm{vec}(\mathbf{B}_{1234}). \tag{3}$$

Where $\mathbf{B}_{12} = \mathrm{mat}_c(\mathcal{B}_{12})^T$, and similarly for $\mathbf{B}_{34}$.

Goal: Refold the solution to build our algorithm

Refolding the solution locally

Based on equation (2),

$$\operatorname{vec}(\mathcal{U}) = (\mathbf{V}_4 \otimes \mathbf{V}_3 \otimes \mathbf{V}_2 \otimes \mathbf{V}_1)(\mathbf{B}_{34} \otimes \mathbf{B}_{12}) \operatorname{vec}(\mathbf{B}_{1234}). \tag{3}$$

Where $\mathbf{B}_{12} = \operatorname{mat}_c(\mathcal{B}_{12})^T$, and similarly for $\mathbf{B}_{34}$.

Goal: Refold the solution to build our algorithm

Define the subtree bases

$$\mathbf{V}_{12}(t) = \big(\mathbf{V}_2(t) \otimes \mathbf{V}_1(t)\big)\mathbf{B}_{12}(t), \tag{4}$$

$$\mathbf{V}_{34}(t) = \big(\mathbf{V}_4(t) \otimes \mathbf{V}_3(t)\big)\mathbf{B}_{34}(t). \tag{5}$$

Define

$$\widetilde{\mathbf{B}}_{12}(t) = \mathbf{B}_{12}(t)\mathbf{B}_{1234}(t), \qquad \widetilde{\mathcal{B}}_{12}(t) \in \mathbb{R}^{r_1(t) \times r_2(t) \times r_{34}(t)}.$$

Refolding as a Tucker tensor

$$\mathcal{U}(t) = \widetilde{\mathcal{B}}_{12}(t) \times_a \mathbf{V}_1(t) \times_b \mathbf{V}_2(t) \times_c \mathbf{V}_{34}(t). \tag{6}$$

- a, b, c label local tensor modes, not fixed dimensions
- For $\mathcal{B}_{34}$, modes a and b instead connect to $\mathbf{V}_3$ and $\mathbf{V}_4$.

Outline

1. Motivation and hierarchical Tucker representation
2. The first-order HT-RAIL algorithm
3. High-order time integration
4. Numerical experiments
5. Closing remarks

The semi-discrete formulation

Let

$$\mathbf{F}_i(t) = d_i(t)\mathbf{D}_{x_i x_i}, \qquad i = 1, 2, 3, 4,$$

where $\mathbf{D}_{x_i x_i}$ is the a spectral differentiation matrix.
Let our spatially discretized diffusion equation be

$$\frac{d\mathcal{U}}{dt} = \mathcal{L}\big(t, \mathcal{U}(t)\big),$$

The semi-discrete formulation

Let

$$\mathbf{F}_i(t) = d_i(t)\mathbf{D}_{x_i x_i}, \qquad i = 1, 2, 3, 4,$$

where $\mathbf{D}_{x_i x_i}$ is the a spectral differentiation matrix.
Let our spatially discretized diffusion equation be

$$\frac{d\mathcal{U}}{dt} = \mathcal{L}\big(t, \mathcal{U}(t)\big),$$

with

$$\begin{aligned}
\mathcal{L}\big(t, \mathcal{U}(t)\big) = \widetilde{\mathcal{B}}_{12}(t) \times_a \big(\mathbf{F}_1(t)\mathbf{V}_1(t)\big) \times_b \mathbf{V}_2(t) \times_c \mathbf{V}_{34}(t) \\
+ \widetilde{\mathcal{B}}_{12}(t) \times_a \mathbf{V}_1(t) \times_b \big(\mathbf{F}_2(t)\mathbf{V}_2(t)\big) \times_c \mathbf{V}_{34}(t) \\
+ \widetilde{\mathcal{B}}_{12}(t) \times_a \mathbf{V}_1(t) \times_b \mathbf{V}_2(t) \times_c \Big[\big(\mathbf{V}_4(t) \otimes \mathbf{F}_3(t)\mathbf{V}_3(t)\big)\mathbf{B}_{34}(t)\Big] \\
+ \widetilde{\mathcal{B}}_{12}(t) \times_a \mathbf{V}_1(t) \times_b \mathbf{V}_2(t) \times_c \Big[\big(\mathbf{F}_4(t)\mathbf{V}_4(t) \otimes \mathbf{V}_3(t)\big)\mathbf{B}_{34}(t)\Big].
\end{aligned} \tag{7}$$

- Each differentiation matrix acts on its corresponding leaf basis.
- Kronecker expressions are evaluated through structured contractions, not assembled as large matrices.
- $\mathcal{L}$ denotes the discrete Laplacian.

K-steps: projecting onto the other modes

Backward Euler gives

$$\mathcal{U}^{n+1} - \Delta t\, \mathcal{L}\left(t^{n+1}, \mathcal{U}^{n+1}\right) = \mathcal{U}^{n}. \tag{8}$$

Goal: Update the leaf basis $\mathbf{V}_1^{n+1}$.

- Flatten in the local mode a.
- Project onto $\mathbf{V}_{34}^{\star,n+1} \otimes \mathbf{V}_2^{\star,n+1}$.

For backward Euler, we use approximations at t^{n+1}:

$$\mathbf{V}_i^{\star,n+1} = \mathbf{V}_i^n, \qquad \mathbf{B}_{34}^{\star,n+1} = \mathbf{B}_{34}^n,$$

$$\mathbf{V}_{34}^{\star,n+1} = \left(\mathbf{V}_4^{\star,n+1} \otimes \mathbf{V}_3^{\star,n+1}\right)\mathbf{B}_{34}^{\star,n+1}.$$

K-steps: projecting onto the other modes

Backward Euler gives

$$\mathcal{U}^{n+1} - \Delta t\, \mathcal{L}\left(t^{n+1}, \mathcal{U}^{n+1}\right) = \mathcal{U}^{n}. \tag{8}$$

Goal: Update the leaf basis $\mathbf{V}_1^{n+1}$.

- Flatten in the local mode a.
- Project onto $\mathbf{V}_{34}^{\star,n+1} \otimes \mathbf{V}_2^{\star,n+1}$.

For backward Euler, we use approximations at t^{n+1}:

$$\mathbf{V}_i^{\star,n+1} = \mathbf{V}_i^{n}, \qquad \mathbf{B}_{34}^{\star,n+1} = \mathbf{B}_{34}^{n},$$

$$\mathbf{V}_{34}^{\star,n+1} = \left(\mathbf{V}_4^{\star,n+1} \otimes \mathbf{V}_3^{\star,n+1}\right)\mathbf{B}_{34}^{\star,n+1}.$$

The projected solution is

$$\mathbf{K}_1^{n+1} := \mathbf{V}_1^{n+1}\, \mathrm{mat}_a\left(\widetilde{\mathcal{B}}_{12}^{n+1}\right)\left(\left(\mathbf{V}_{34}^{n+1}\right)^T \mathbf{V}_{34}^{\star,n+1} \otimes \left(\mathbf{V}_2^{n+1}\right)^T \mathbf{V}_2^{\star,n+1}\right). \tag{9}$$

K-steps: a Sylvester equation

Use orthonormal predicted bases and a Galerkin projection onto approximated future bases

$$
\begin{aligned}
(\mathbf{I} - \Delta t \mathbf{F}_1)\mathbf{K}_1^{n+1} - \Delta t\, \mathbf{K}_1^{n+1} \Bigg\{ \Big[(\mathbf{B}_{34}^{\star,n+1})^T \Big((\mathbf{V}_4^{\star,n+1})^T \mathbf{F}_4 \mathbf{V}_4^{\star,n+1} \otimes \mathbf{I} \\
+ \mathbf{I} \otimes (\mathbf{V}_3^{\star,n+1})^T \mathbf{F}_3 \mathbf{V}_3^{\star,n+1} \Big) \mathbf{B}_{34}^{\star,n+1} \Big] \otimes \mathbf{I} \\
+ \mathbf{I} \otimes (\mathbf{V}_2^{\star,n+1})^T \mathbf{F}_2 \mathbf{V}_2^{\star,n+1} \Bigg\}^T = \mathbf{K}_1^n
\end{aligned}
\tag{10}
$$

K-steps: a Sylvester equation

Use orthonormal predicted bases and a Galerkin projection onto approximated future bases

$$
\begin{aligned}
(\mathbf{I} - \Delta t \mathbf{F}_1)\mathbf{K}_1^{n+1} - \Delta t\, \mathbf{K}_1^{n+1} \Bigg\{ \Big[(\mathbf{B}_{34}^{\star,n+1})^T \Big((\mathbf{V}_4^{\star,n+1})^T \mathbf{F}_4 \mathbf{V}_4^{\star,n+1} \otimes \mathbf{I} \\
+ \mathbf{I} \otimes (\mathbf{V}_3^{\star,n+1})^T \mathbf{F}_3 \mathbf{V}_3^{\star,n+1} \Big) \mathbf{B}_{34}^{\star,n+1} \Big] \otimes \mathbf{I} \\
+ \mathbf{I} \otimes (\mathbf{V}_2^{\star,n+1})^T \mathbf{F}_2 \mathbf{V}_2^{\star,n+1} \Bigg\}^T = \mathbf{K}_1^n
\end{aligned}
\tag{10}
$$

- A Sylvester equation: $\mathbf{A}\mathbf{K}_1^{n+1} + \mathbf{K}_1^{n+1}\mathbf{B} = \mathbf{C}$.
- Solve using MATLAB's `sylvester`.
- Repeat for the other leaf bases $\mathbf{V}_2, \mathbf{V}_3, \mathbf{V}_4$.

Identity matrix sizes follow from the adjacent factors.

Extracting and enriching the leaf bases

Extract the updated basis using a reduced QR factorization:

$$\mathbf{K}_i^{n+1} = \mathbf{V}_i^{\ddagger,n+1}\mathbf{R}_i^{n+1}, \qquad i = 1, 2, 3, 4. \tag{11}$$

Extracting and enriching the leaf bases

Extract the updated basis using a reduced QR factorization:

$$\mathbf{K}_i^{n+1} = \mathbf{V}_i^{\ddagger,n+1}\mathbf{R}_i^{n+1}, \qquad i = 1, 2, 3, 4. \tag{11}$$

Reduced augmentation: retain new and previous basis information.

$$\left[\mathbf{V}_i^{\ddagger,n+1},\ \mathbf{V}_i^{n}\right] = \mathbf{Q}_i^{n+1}\mathbf{R}_{i,\mathrm{aug}}^{n+1}. \tag{12}$$

Retaining left singular vectors with $\sigma_j > 10^{-12}$ (stored in $\widehat{\mathbf{Y}}_i^{n+1}$) gives

$$\widehat{\mathbf{V}}_i^{n+1} = \mathbf{Q}_i^{n+1}\widehat{\mathbf{Y}}_i^{n+1}. \tag{13}$$

Extracting and enriching the leaf bases

Extract the updated basis using a reduced QR factorization:

$$\mathbf{K}_i^{n+1} = \mathbf{V}_i^{\ddagger,n+1}\mathbf{R}_i^{n+1}, \qquad i = 1,2,3,4. \tag{11}$$

Reduced augmentation: retain new and previous basis information.

$$\left[\mathbf{V}_i^{\ddagger,n+1},\ \mathbf{V}_i^{n}\right] = \mathbf{Q}_i^{n+1}\mathbf{R}_{i,\mathrm{aug}}^{n+1}. \tag{12}$$

Retaining left singular vectors with $\sigma_j > 10^{-12}$ (stored in $\widehat{\mathbf{Y}}_i^{n+1}$) gives

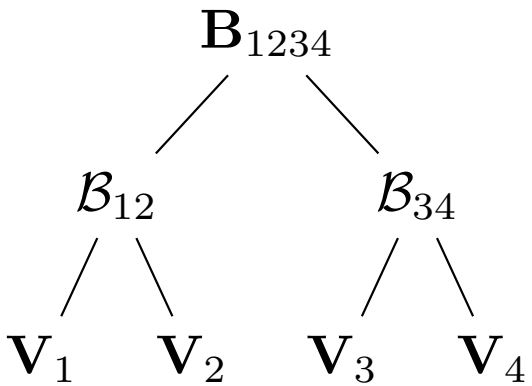
$$\widehat{\mathbf{V}}_i^{n+1} = \mathbf{Q}_i^{n+1}\widehat{\mathbf{Y}}_i^{n+1}. \tag{13}$$

- Retain information from both ends of the time interval.
- Remove redundant directions introduced by concatenation.
- Use the enriched bases in the core updates.

$\star$ prediction; $\ddagger$ K-step basis; $\widehat{\ }$ enriched, before final truncation.

K-steps: Concluding

Recall: our solution is stored in the HT tree

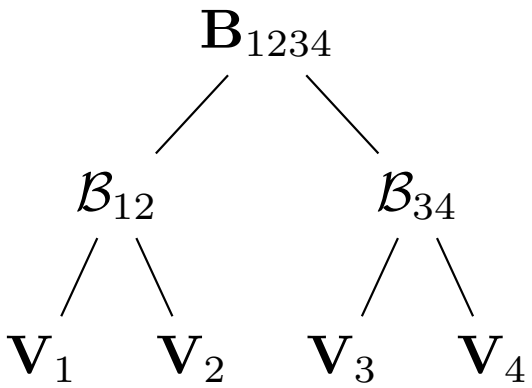


$$\mathcal{U} = (\mathbf{V}_{34} \otimes \mathbf{V}_2 \otimes \mathbf{V}_1)\,\mathrm{vec}(\tilde{\mathcal{B}}_{12})$$

- **Know:** Updated, reduced augmented leaf bases $\mathbf{V}_i, i \in \{1, 2, 3, 4\}$
- **Next:** Update the intermediate core tensors $\mathcal{B}_{12}$ and $\mathcal{B}_{34}$

K-steps: Concluding

Recall: our solution is stored in the HT tree



$$\mathcal{U} = (\mathbf{V}_{34} \otimes \mathbf{V}_2 \otimes \mathbf{V}_1)\,\mathrm{vec}(\tilde{\mathcal{B}}_{12})$$

- **Know:** Updated, reduced augmented leaf bases $\mathbf{V}_i, i \in \{1,2,3,4\}$
- **Next:** Update the intermediate core tensors $\mathcal{B}_{12}$ and $\mathcal{B}_{34}$

For the $\{1,2\}$ node, project onto

$$\mathbf{V}_{34}^{\star,n+1} \otimes \widehat{\mathbf{V}}_2^{n+1} \otimes \widehat{\mathbf{V}}_1^{n+1}.$$

Define the projected solution for the intermediate core

$$\mathrm{vec}(\hat{\tilde{\mathcal{B}}}_{12}^{n+1}) \;:=\; \left((\mathbf{V}_{34}^{\star,n+1})^T \mathbf{V}_{34}^{n+1} \otimes (\hat{\mathbf{V}}_2^{n+1})^T \mathbf{V}_2^{n+1} \otimes (\hat{\mathbf{V}}_1^{n+1})^T \mathbf{V}_1^{n+1}\right) \mathrm{vec}(\tilde{\mathcal{B}}_{12}^{n+1}).$$

B-steps: updating the intermediate cores

Projecting our update equation yields

$$
\begin{aligned}
\Big\{ & (\mathbf{B}_{34}^{\star,n+1})^T \Big(-\Delta t (\hat{\mathbf{V}}_4^{n+1})^T (\mathbf{F}_4 \hat{\mathbf{V}}_4^{n+1}) \otimes \mathbf{I}_{\hat{r}_3^{n+1}} \Big) \mathbf{B}_{34}^{\star,n+1} \otimes \mathbf{I}_{\hat{r}_2^{n+1}} \otimes \mathbf{I}_{\hat{r}_1^{n+1}} \\
& + (\mathbf{B}_{34}^{\star,n+1})^T \Big(\mathbf{I}_{\hat{r}_4^{n+1}} \otimes -\Delta t (\hat{\mathbf{V}}_4^{n+1})^T (\mathbf{F}_4 \hat{\mathbf{V}}_4^{n+1}) \Big) \mathbf{B}_{34}^{\star,n+1} \otimes \mathbf{I}_{\hat{r}_2^{n+1}} \otimes \mathbf{I}_{\hat{r}_1^{n+1}} \\
& + \mathbf{I}_{\hat{r}_{34}^{n+1}} \otimes \Big(-\Delta t (\hat{\mathbf{V}}_2^{n+1})^T (\mathbf{F}_2 \hat{\mathbf{V}}_2^{n+1}) \Big) \otimes \mathbf{I}_{\hat{r}_1^{n+1}} \\
& + \mathbf{I}_{\hat{r}_{34}^{n+1}} \otimes \mathbf{I}_{\hat{r}_2^{n+1}} \otimes \Big(-\Delta t (\hat{\mathbf{V}}_1^{n+1})^T (\mathbf{F}_1 \hat{\mathbf{V}}_1^{n+1}) \Big) \Big\} \operatorname{vec}(\hat{\tilde{\mathcal{B}}}_{12}^{n+1}) = \operatorname{vec}(\hat{\tilde{\mathcal{B}}}_{12}^{n}).
\end{aligned}
\tag{14}
$$

- Solve using the tensor direct solver based on Simoncini's approach.[1]
- Reshape and extract the updated intermediate core: (Repeat on the right subtree for $\widehat{\mathbf{B}}_{34}^{n+1}$)

B-steps: updating the root and truncating

Build the updated subtree bases:

$$\widehat{\mathbf{V}}_{12}^{n+1} = \big(\widehat{\mathbf{V}}_{2}^{n+1} \otimes \widehat{\mathbf{V}}_{1}^{n+1}\big)\widehat{\mathbf{B}}_{12}^{n+1},$$
$$\widehat{\mathbf{V}}_{34}^{n+1} = \big(\widehat{\mathbf{V}}_{4}^{n+1} \otimes \widehat{\mathbf{V}}_{3}^{n+1}\big)\widehat{\mathbf{B}}_{34}^{n+1}.$$

For concision we define

$$\widehat{\mathbf{F}}_{12}^{n+1} = (\widehat{\mathbf{B}}_{12}^{n+1})^T \Big[(\widehat{\mathbf{V}}_{2}^{n+1})^T \mathbf{F}_2 \widehat{\mathbf{V}}_{2}^{n+1} \oplus (\widehat{\mathbf{V}}_{1}^{n+1})^T \mathbf{F}_1 \widehat{\mathbf{V}}_{1}^{n+1}\Big] \widehat{\mathbf{B}}_{12}^{n+1}, \tag{15}$$

and define $\widehat{\mathbf{F}}_{34}^{n+1}$ analogously on the right subtree.

Projecting onto the two updated subtree bases gives

$$(\mathbf{I} - \Delta t \widehat{\mathbf{F}}_{12}^{n+1})\widehat{\mathbf{B}}_{1234}^{n+1} - \Delta t\, \widehat{\mathbf{B}}_{1234}^{n+1}(\widehat{\mathbf{F}}_{34}^{n+1})^T = (\widehat{\mathbf{V}}_{12}^{n+1})^T \mathbf{V}_{12}^{n} \mathbf{B}_{1234}^{n} (\mathbf{V}_{34}^{n})^T \widehat{\mathbf{V}}_{34}^{n+1}. \tag{16}$$

Finally, apply hierarchical truncation:

$$\mathrm{HT}(\mathcal{U}^{n+1}) = \mathtt{trun}_{\varepsilon}\big(\,\mathrm{HT}(\widehat{\mathcal{U}}^{n+1})\big), \qquad \varepsilon = 10^{-6}.$$

Hierarchical truncation: Kressner–Tobler `htucker` toolbox.

Outline

1. Motivation and hierarchical Tucker representation
2. The first-order HT-RAIL algorithm
3. High-order time integration
4. Numerical experiments
5. Closing remarks

High-order integration with DIRK methods

For an s-stage diagonally implicit Runge–Kutta method,

$$\mathcal{U}^{(k)} = \mathcal{U}^n + \Delta t \sum_{\ell=1}^{k} a_{k\ell} \mathcal{L}(t^{(\ell)}, \mathcal{U}^{(\ell)}), \qquad t^{(k)} = t^n + c_k \Delta t. \tag{17}$$

Separate the known contributions:

$$\mathcal{W}^{(k-1)} = \mathcal{U}^n + \Delta t \sum_{\ell=1}^{k-1} a_{k\ell} \mathcal{L}(t^{(\ell)}, \mathcal{U}^{(\ell)}), \tag{18}$$

$$\boxed{\mathcal{U}^{(k)} - a_{kk} \Delta t \, \mathcal{L}(t^{(k)}, \mathcal{U}^{(k)}) = \mathcal{W}^{(k-1)}.} \tag{19}$$

High-order integration with DIRK methods

For an s-stage diagonally implicit Runge–Kutta method,

$$\mathcal{U}^{(k)} = \mathcal{U}^n + \Delta t \sum_{\ell=1}^{k} a_{k\ell} \mathcal{L}(t^{(\ell)}, \mathcal{U}^{(\ell)}), \qquad t^{(k)} = t^n + c_k \Delta t. \tag{17}$$

Separate the known contributions:

$$\mathcal{W}^{(k-1)} = \mathcal{U}^n + \Delta t \sum_{\ell=1}^{k-1} a_{k\ell} \mathcal{L}(t^{(\ell)}, \mathcal{U}^{(\ell)}), \tag{18}$$

$$\boxed{\mathcal{U}^{(k)} - a_{kk} \Delta t \, \mathcal{L}(t^{(k)}, \mathcal{U}^{(k)}) = \mathcal{W}^{(k-1)}.} \tag{19}$$

- Each stage has the structure of a backward Euler solve.
- In the K- and B-steps, replace Δt by $a_{kk}\Delta t$ and $\mathcal{U}^n$ by $\mathcal{W}^{(k-1)}$.

Stiffly accurate DIRK: $a_{s\ell} = b_\ell$, $c_s = 1$, and $\mathcal{U}^{n+1} = \mathcal{U}^{(s)}$.

Methods tested: backward Euler, DIRK2, and DIRK3.

Enriching the projection spaces at each stage

Goal: Project out bases at updated stages for implicit integration.
Challenge: The basis at the new stage is not yet known.
Construct a first-order prediction $\mathcal{U}^{\dagger,(k)}$ at $t^{(k)}$. Combine it with previous stage solutions:

$$\begin{aligned}\mathrm{HT}(\mathcal{U}^{\star,(k)}) = \mathrm{redaug}\Big(&\mathrm{HT}(\mathcal{U}^{\dagger,(k)}),\ \mathrm{HT}(\mathcal{U}^{(k-1)}),\\ &\ldots,\ \mathrm{HT}(\mathcal{U}^{(1)}),\ \mathrm{HT}(\mathcal{U}^{n})\Big).\end{aligned} \tag{20}$$

- Sum the HT tensors and compress with tolerance ε.
- The predicted projection includes transfer tensors:

$$\mathbf{V}_{34}^{\star,(k)} = (\mathbf{V}_{4}^{\star,(k)} \otimes \mathbf{V}_{3}^{\star,(k)})\mathbf{B}_{34}^{\star,(k)}.$$

Enriching the projection spaces at each stage

Goal: Project out bases at updated stages for implicit integration.
Challenge: The basis at the new stage is not yet known.
Construct a first-order prediction $\mathcal{U}^{\dagger,(k)}$ at $t^{(k)}$. Combine it with previous stage solutions:

$$\begin{aligned}\mathrm{HT}(\mathcal{U}^{\star,(k)}) = \mathrm{redaug}\Big(&\mathrm{HT}(\mathcal{U}^{\dagger,(k)}),\ \mathrm{HT}(\mathcal{U}^{(k-1)}),\\ &\ldots,\ \mathrm{HT}(\mathcal{U}^{(1)}),\ \mathrm{HT}(\mathcal{U}^{n})\Big).\end{aligned} \tag{20}$$

- Sum the HT tensors and compress with tolerance ε.
- The predicted projection includes transfer tensors:

$$\mathbf{V}_{34}^{\star,(k)} = (\mathbf{V}_{4}^{\star,(k)} \otimes \mathbf{V}_{3}^{\star,(k)})\mathbf{B}_{34}^{\star,(k)}.$$

After each K-step, enrich the updated leaf basis:

$$\widehat{\mathbf{V}}_{i}^{(k)} = \mathrm{redaug}\left[\mathbf{V}_{i}^{\ddagger,(k)},\ \mathbf{V}_{i}^{(k-1)},\ldots,\mathbf{V}_{i}^{(1)},\mathbf{V}_{i}^{n}\right]. \tag{21}$$

- Perform the intermediate-core and root updates.
- Truncate the resulting HT solution at each stage.

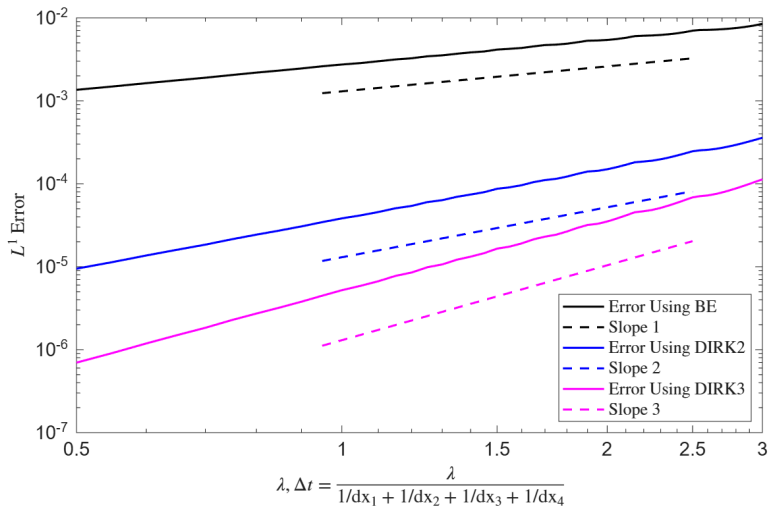
Outline

1. Motivation and hierarchical Tucker representation
2. The first-order HT-RAIL algorithm
3. High-order time integration
4. Numerical experiments
5. Closing remarks

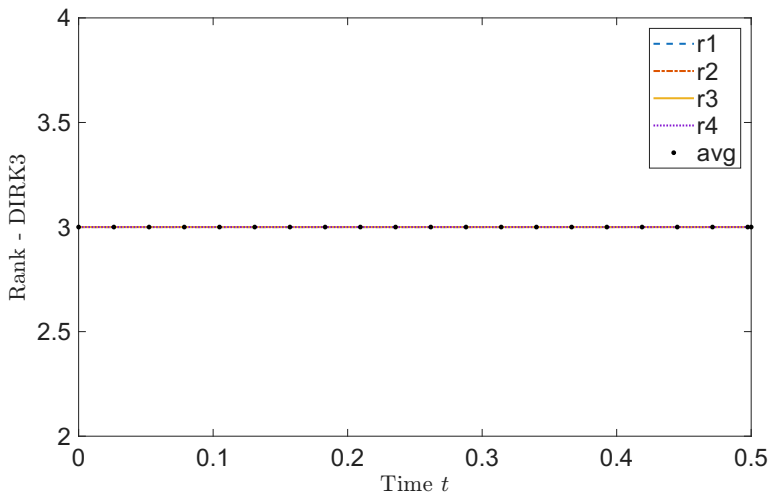
Accuracy test: the first three Fourier modes

$$u_t = 0.1 \sum_{i=1}^{4} u_{x_i x_i}, \qquad u_0(\mathbf{x}) = \sum_{k=1}^{3} \prod_{i=1}^{4} \sin(k x_i).$$

$$\Omega = [0, 2\pi]^4, \qquad N_i = 60, \qquad T_f = 0.5, \qquad \Delta t = \lambda \Delta x / 4.$$



Normalized L^1 error: $\|u - u_h\|_{L^1} / |\Omega|$.



Rank history for DIRK3.

- Observed temporal orders: 1, 2, and 3.
- The reported solution remains rank 3 in each direction.

Rank tests: setup

Initial Condition: Use a sum of three separable Gaussian functions:

$$u_0(\mathbf{x}) = 0.8\, e^{-15\|\mathbf{x}-6.5\|^2} + 0.5\, e^{-15\|\mathbf{x}-7.5\|^2} + 1.2\, e^{-15\|\mathbf{x}-4.5\|^2}. \tag{22}$$

$$\Omega = [0, 14]^4, \qquad N_i = 60, \qquad T_f = 10, \qquad \lambda = 1, \qquad \Delta t = \Delta x/4.$$

- Periodic boundary conditions and spectral differentiation.
- Compare backward Euler, DIRK2, and DIRK3.
- Vary the diffusion coefficients for

$$u_t = \sum_{i=1}^{d} d_i(t) u_{x_i x_i}$$

Rank tests: setup

Initial Condition: Use a sum of three separable Gaussian functions:

$$u_0(\mathbf{x}) = 0.8\, e^{-15\|\mathbf{x}-6.5\|^2} + 0.5\, e^{-15\|\mathbf{x}-7.5\|^2} + 1.2\, e^{-15\|\mathbf{x}-4.5\|^2}. \tag{22}$$

$$\Omega = [0, 14]^4, \qquad N_i = 60, \qquad T_f = 10, \qquad \lambda = 1, \qquad \Delta t = \Delta x/4.$$

- Periodic boundary conditions and spectral differentiation.
- Compare backward Euler, DIRK2, and DIRK3.
- Vary the diffusion coefficients for

$$u_t = \sum_{i=1}^{d} d_i(t) u_{x_i x_i}$$

Track two types of ranks:

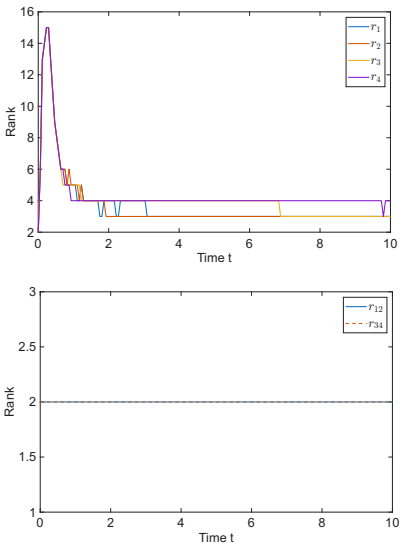
- Leaf ranks: r_1, r_2, r_3, r_4.
- Intermediate ranks: r_{12}, r_{34}.

These are ranks of the computed, truncated solution; they depend on the time integrator, projection spaces, and tolerance.

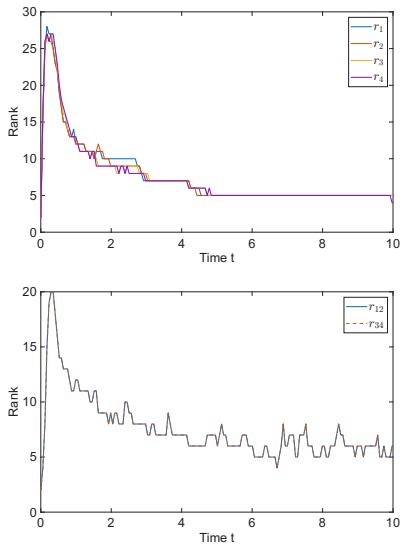
Rank test #1: constant coefficients

$$d_1(t) = d_2(t) = d_3(t) = d_4(t) = 1.$$

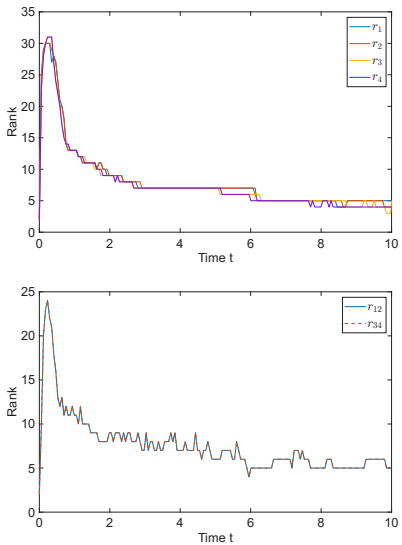
Backward Euler



DIRK2



DIRK3



Top: leaf ranks r_1, r_2, r_3, r_4. Bottom: intermediate ranks r_{12}, r_{34}.

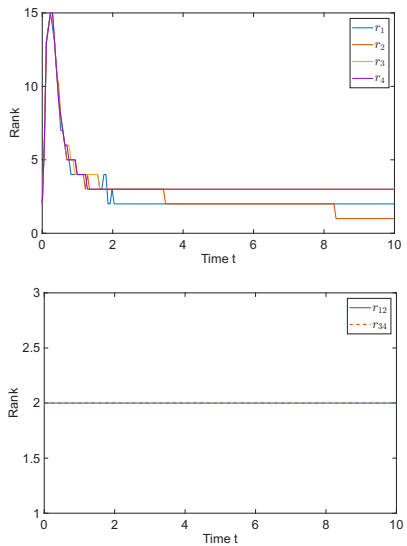
- The reported leaf ranks initially increase and then decrease.
- Backward Euler exhibits a faster decrease in leaf ranks.
- Its intermediate ranks remain at 2 in the reported experiment.

Takeaway: Stage enrichment substantially changes the computed rank histories.

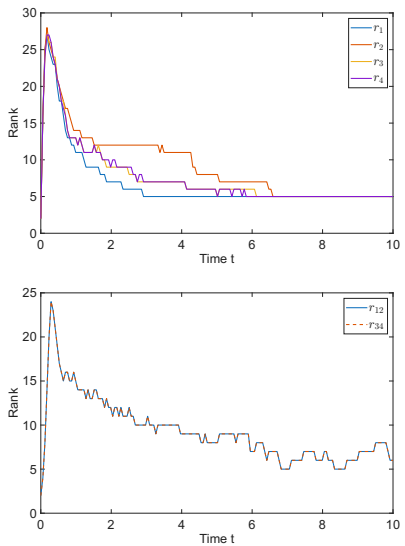
Rank test #2: sinusoidal coefficients

$$d_1(t) = 1 + \sin(2\pi t/T_f), \quad d_2(t) = 1 - \sin(2\pi t/T_f),$$
$$d_3(t) = d_4(t) = 1$$

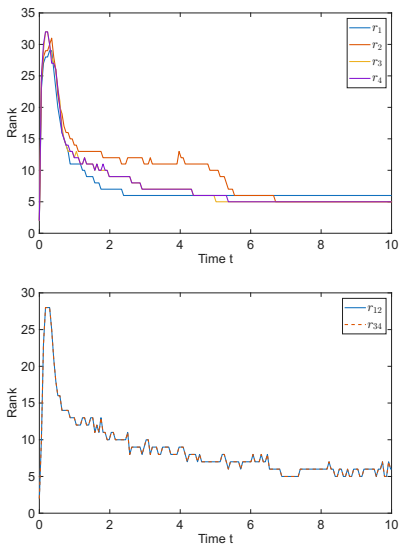
Backward Euler



DIRK2



DIRK3



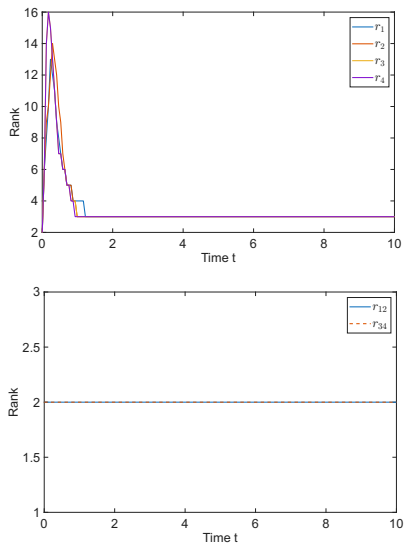
Top: leaf ranks r_1, r_2, r_3, r_4. Bottom: intermediate ranks r_{12}, r_{34}.

- DIRK2 and DIRK3 show distinct histories for r_1 and r_2.
- r_2 has a plateau near $t = T_f/4 = 2.5$, where $d_2(t) = 0$.
- Backward Euler does not show the same rank variation.

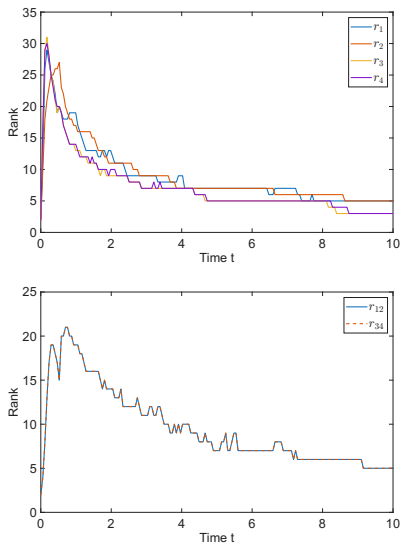
Rank test #3: square-wave coefficients

$$d_1(t) = 1.05 + \tfrac{1}{2}\,\mathrm{sign}(\sin(2\pi t)),$$
$$d_2(t) = 1.05 - \tfrac{1}{2}\,\mathrm{sign}(\sin(2\pi t)), \qquad d_3(t) = d_4(t) = 1.$$

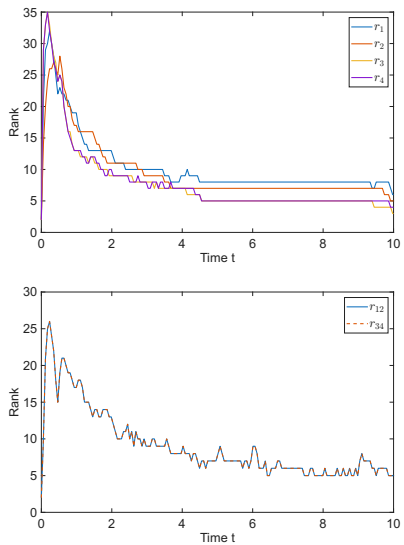
Backward Euler



DIRK2



DIRK3



Top: leaf ranks r_1, r_2, r_3, r_4. Bottom: intermediate ranks r_{12}, r_{34}.

- The numerical rank histories respond to alternating diffusion rates.
- The manuscript reports a lower peak for r_2 than for r_1.
- Differences between backward Euler and the higher-order methods persist.

Outline

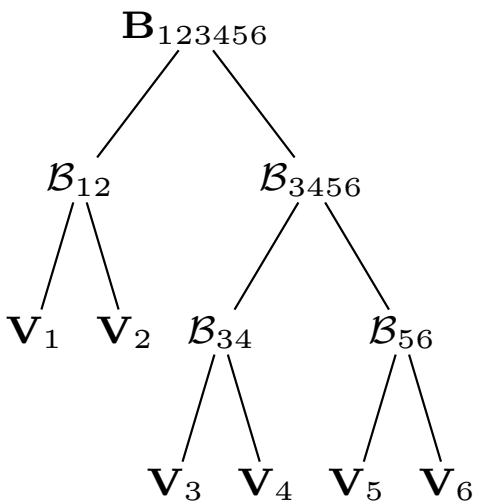
1. Motivation and hierarchical Tucker representation
2. The first-order HT-RAIL algorithm
3. High-order time integration
4. Numerical experiments
5. Closing remarks

Closing remarks

At each DIRK stage:

1. Predict and enrich the HT projection spaces.
2. Solve K-steps for the leaf bases.
3. Extract and augment the updated bases.
4. Solve B-steps for intermediate cores.
5. Solve the root Sylvester equation.
6. Truncate the HT solution.

Extension to $d = 6$

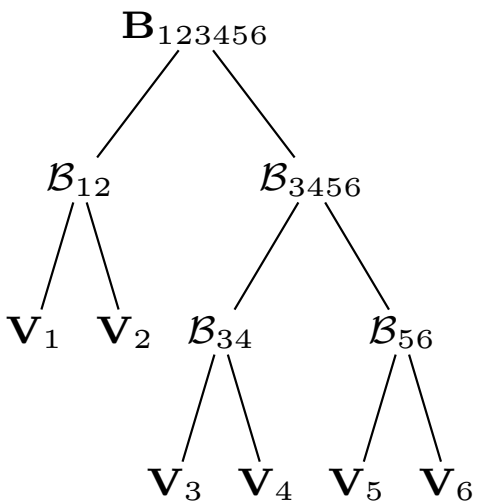


Closing remarks

At each DIRK stage:

1. Predict and enrich the HT projection spaces.
2. Solve K-steps for the leaf bases.
3. Extract and augment the updated bases.
4. Solve B-steps for intermediate cores.
5. Solve the root Sylvester equation.
6. Truncate the HT solution.

Extension to $d = 6$



- The same local projection strategy applies recursively.
- Updated child bases enter their parent's reduced equation.
- Higher-dimensional extension preserves the leaf-to-root structure.

Closing remarks

Recap:

- A high-order rank-adaptive implicit integrator in hierarchical Tucker format.
- 2nd- and 3rd-order accuracy in time
- Rank captures physical changes for DIRK2 and DIRK3

Ongoing work:

- Extend to advection–diffusion equations using IMEX integration.
- Apply the leaf-to-root construction in higher dimensions.

Closing remarks

Recap:

- A high-order rank-adaptive implicit integrator in hierarchical Tucker format.
- 2nd- and 3rd-order accuracy in time
- Rank captures physical changes for DIRK2 and DIRK3

Ongoing work:

- Extend to advection–diffusion equations using IMEX integration.
- Apply the leaf-to-root construction in higher dimensions.

Thank you. Questions?